Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of global renewable energy trends across primary energy and electri

Audience: Policy analysts and energy sector strategists

Source: Our World in Data (compiled in source brief)

75% of Global GHG Emissions Still Come from Fossil Fuels

Fossil fuels have dominated the global energy system since the Industrial Revolution.

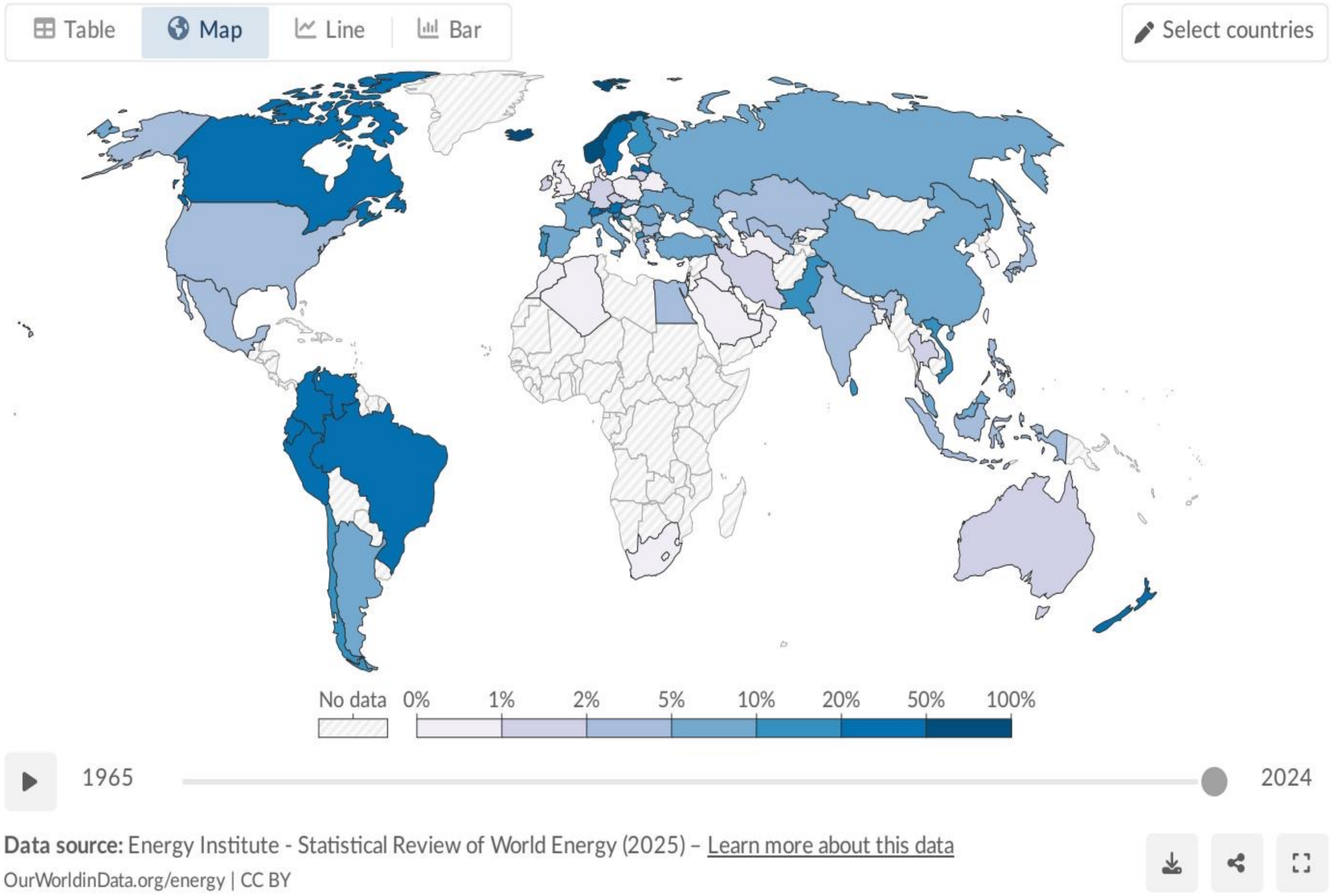
Around three-quarters of global greenhouse gas emissions are linked to fossil energy use.

Air pollution from fossil fuel combustion contributes to more than 5 million premature deaths annually.

The renewable transition is not only a climate priority but also a public-health and economic resilience imperative.

Source: Our World in Data (compiled in source brief)

Renewables Now Supply ~14% of Global Primary Energy



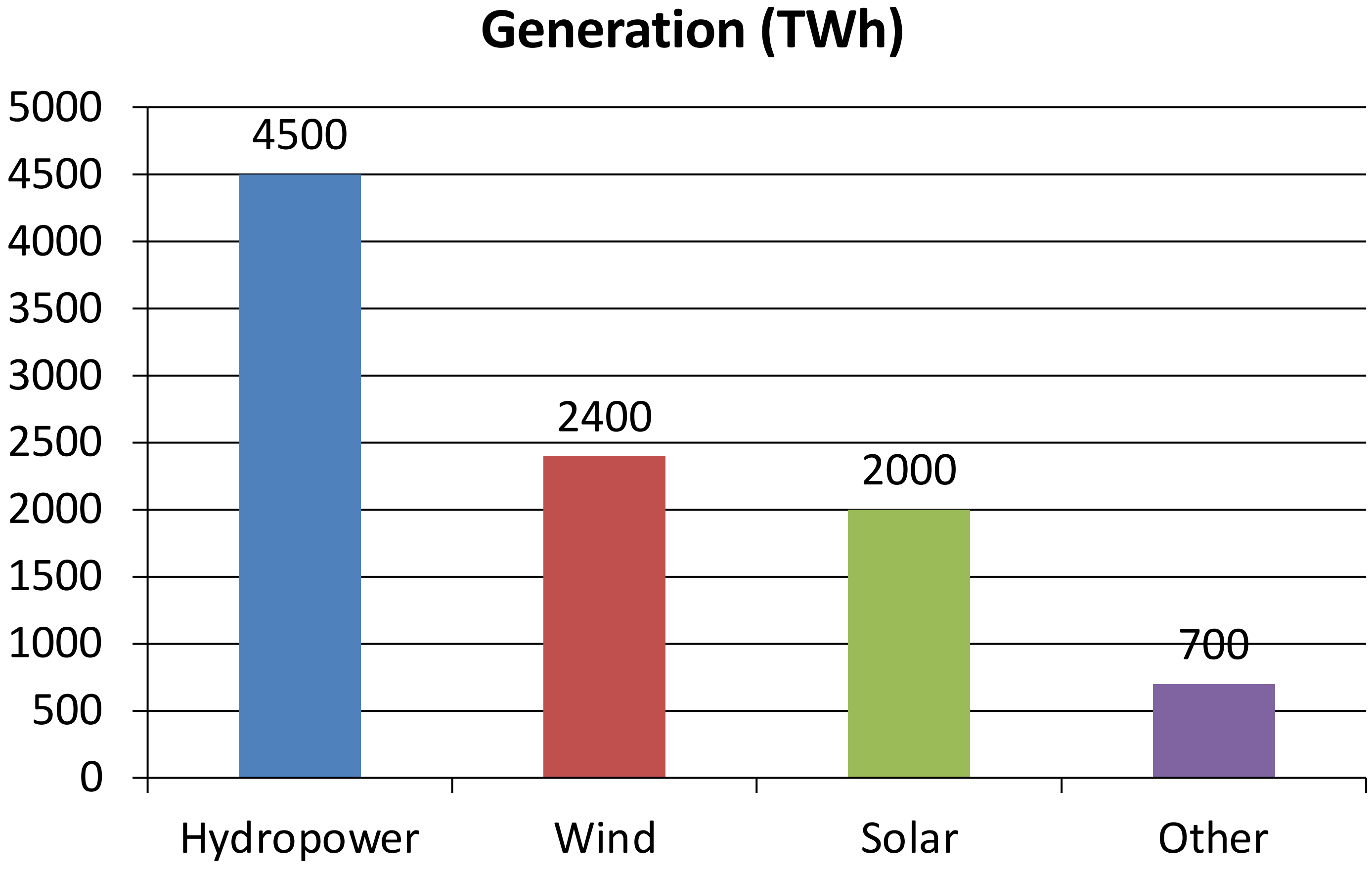
As of 2024, renewables account for about one-seventh of global primary energy.

Primary energy is reported using the substitution method in the source.

This share remains much lower than electricity because transport and heating still rely heavily on fossil fuels.

Source: Our World in Data (compiled in source brief)

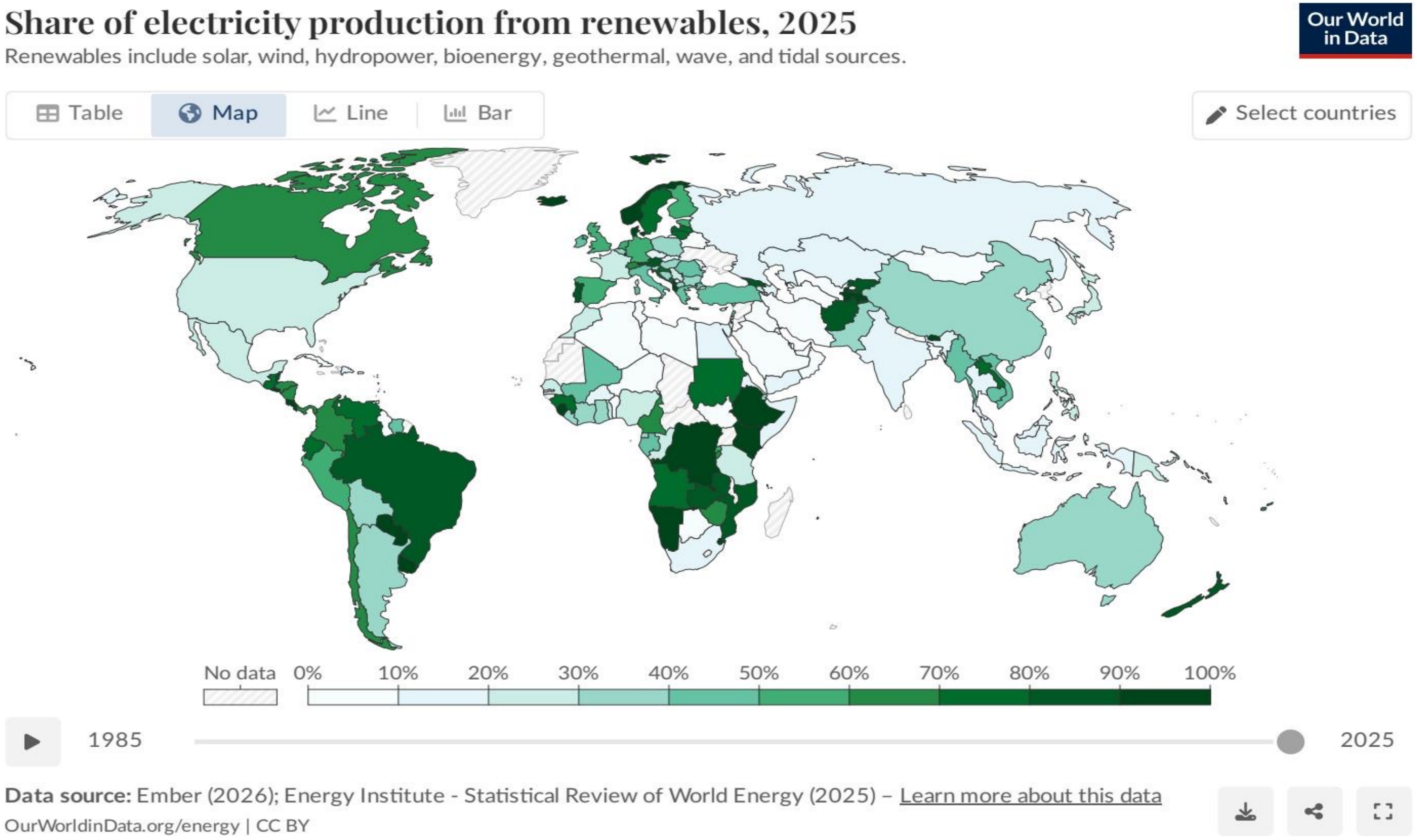
Renewable Electricity Generation Is Growing Rapidly, Led by Wind and So



Hydropower remains the largest source (~4,500 TWh).
Wind (~2,400 TWh) and solar (~2,000 TWh) have acceler
Combined wind+solar output (~4,400 TWh) now nearly r
Other renewables contribute ~700 TWh.

Source: Our World in Data (compiled in source brief)

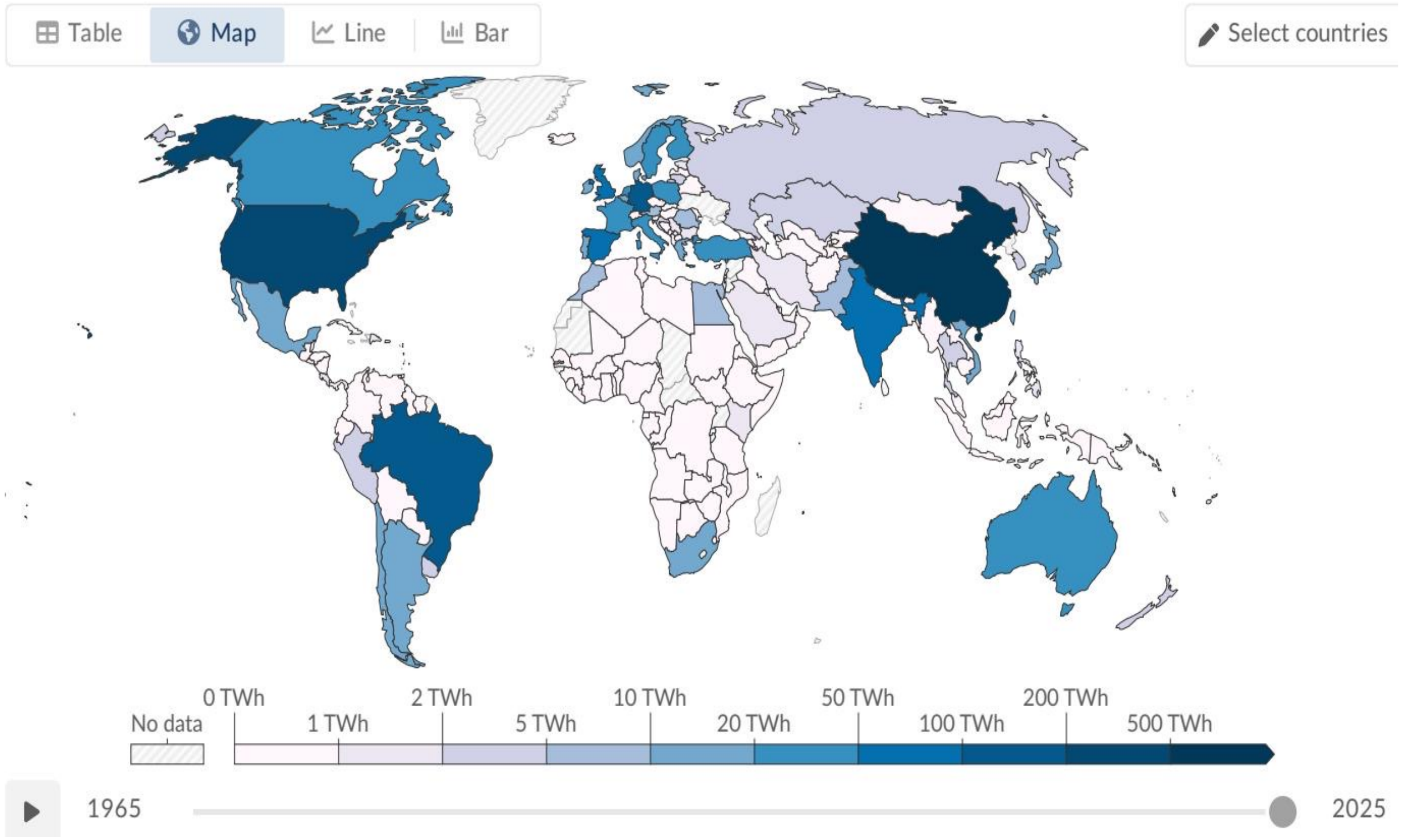
Almost One-Third of Global Electricity Now Comes from Renewables



Country/Region	Renewable elec. share
Brazil	~89%
Canada	~67%
Norway	~98%
China	~31%
India	~22%
Saudi Arabia	~1%

Source: Our World in Data (compiled in source brief)

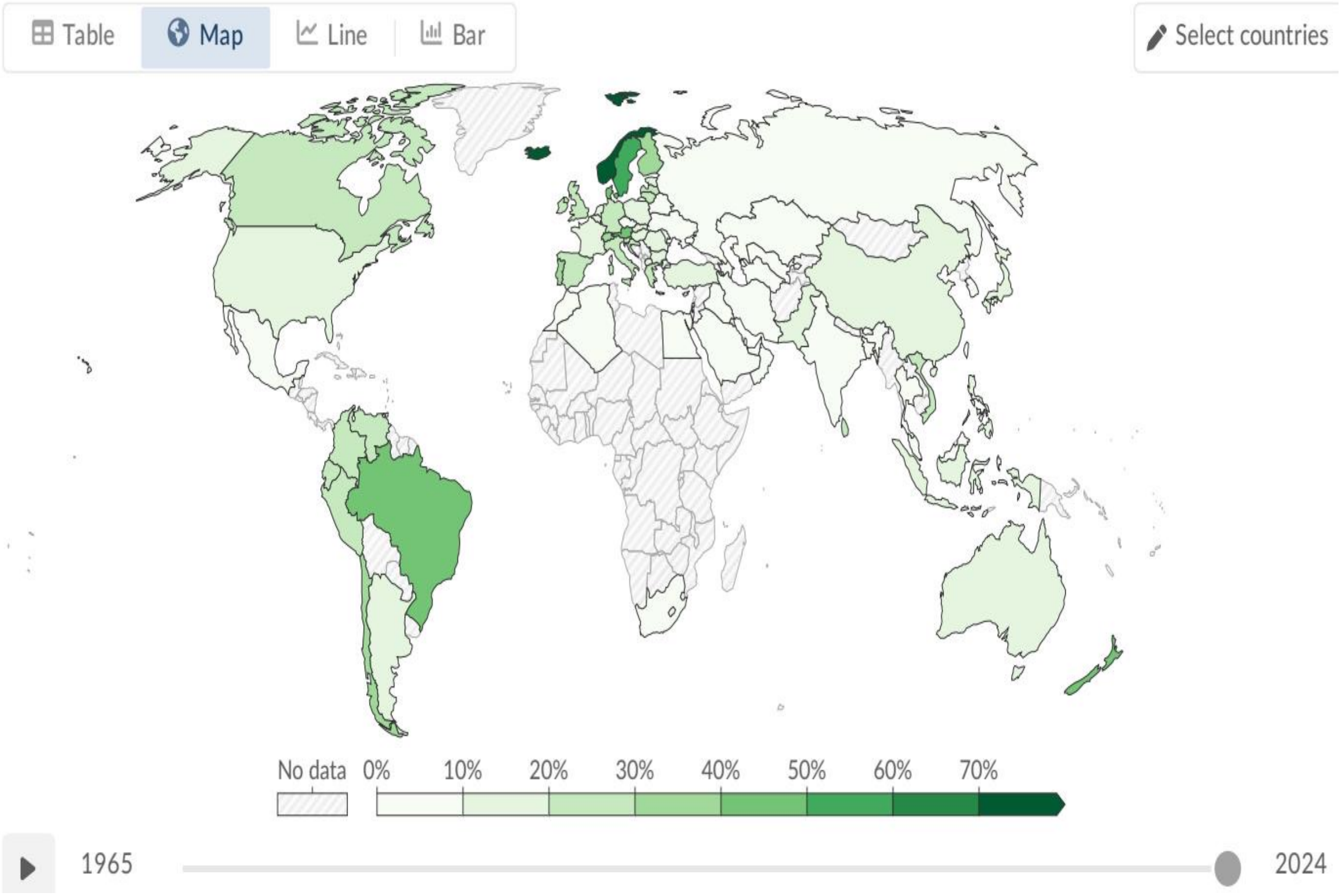
Hydropower Remains the Largest Renewable Source at ~50% of Generatio

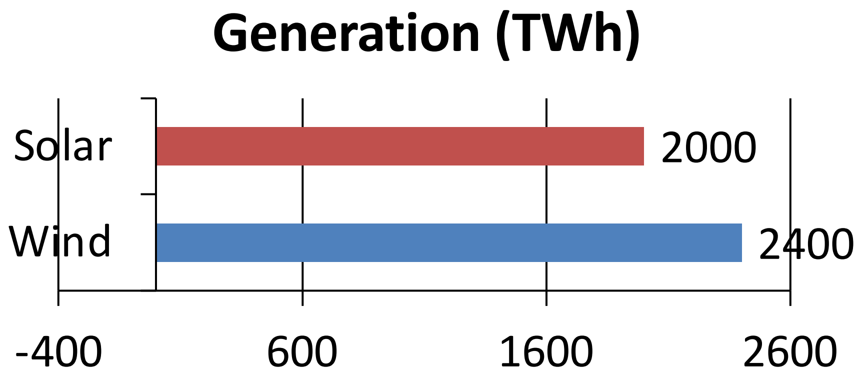


Hydropower provides roughly half of global rene
Top producers include China, Brazil, and Canada.
Hydro remains foundational, but future growth i

Source: Our World in Data (compiled in source brief)

Wind and Solar Are the Fastest-Growing Energy Technologies in History

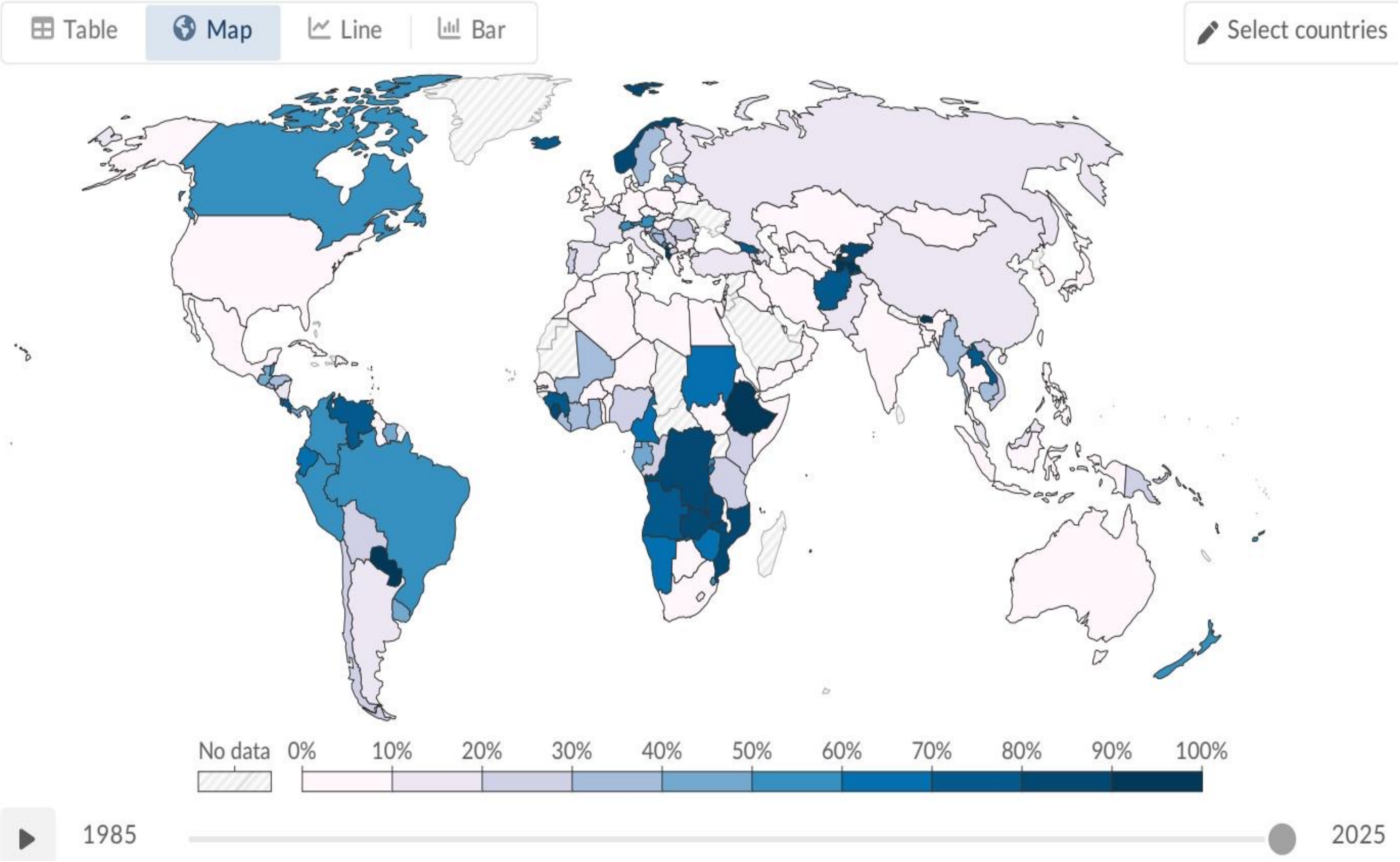




Wind: ~2,400 TWh (~25% of renewable electricity)
Solar: ~2,000 TWh (~21% of renewable electricity)
Solar has seen the sharpest post-2010 acceleration due to cost decline
Together, wind + solar now nearly match hydropower output.

Source: Our World in Data (compiled in source brief)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin



Scandinavia and parts of Latin America achieve very high re
Many Middle Eastern and Sub-Saharan African countries re
Resource endowments and policy investment explain much
Uneven infrastructure deployment is now a central transiti

Source: Our World in Data (compiled in source brief)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Acc

Renewables now provide ~14% of global primary energy and ~30% of global electricity.

Hydropower still leads (~50% of renewable electricity), while wind and solar are rapidly closing the gap.

Regional disparities are large, with leaders above 60–90% renewable electricity and laggards below 10%.

Electricity is advancing faster than transport and heating, which keeps total primary-energy progress slower.

Policy priority for 2025–2030: scale grids, storage, and financing to accelerate deployment where progress lags.

Source: Our World in Data (compiled in source brief)